

Computer Networking

Chapter 1: Computer Networks and the Internet

Questions & Answers

Question 1

We have a 1 Gb/s link. Each user requires 100 Mb/s when active and is active 10% of the time.

1. Under **circuit switching**, how many users can the link support?
2. Under **packet switching**, with 35 users, what is the probability that more than 10 users are active at the same time?

Answer

(a) **Circuit Switching:**

$$\text{Max Users} = \frac{1000 \text{ Mb/s}}{100 \text{ Mb/s}} = 10$$

Hence, only **10** users can be supported.

(b) **Packet Switching:** In packet switching, bandwidth is shared dynamically. Each user is active with probability $p = 0.1$ (10%) and inactive with probability 0.9. Let X = number of active users. Then:

$$X \sim \text{Binomial}(n = 35, p = 0.1).$$

We want the probability that more than 10 users are active at the same time:

$$P(X > 10) = 1 - P(X \leq 10).$$

Expanding:

$$P(X > 10) = 1 - \sum_{k=0}^{10} \binom{35}{k} (0.1)^k (0.9)^{35-k}.$$

Answer (numerical): Evaluating the sum (using exact binomial computation in software or a scientific calculator) gives

$$P(X > 10) \approx 0.0004243 \approx 4.24 \times 10^{-4}.$$

Interpretation: This means that with 35 users who are each active 10% of the time, the chance that more than 10 users are simultaneously active is about 0.042%, i.e. extremely small. —

Question 2

A packet of length $L = 10$ Kbits is transmitted over a link with transmission rate $R = 100$ Mbps.

1. Calculate the one-hop transmission delay.
2. Express your answer in milliseconds.

Answer

Step 1: Formula

$$d_{\text{trans}} = \frac{L}{R}$$

Step 2: Substitution

$$d_{\text{trans}} = \frac{10 \times 10^3 \text{ bits}}{100 \times 10^6 \text{ bits/sec}}$$

Step 3: Simplify

$$d_{\text{trans}} = \frac{10^4}{10^8} = 10^{-4} \text{ sec}$$

Step 4: Convert to ms

$$d_{\text{trans}} = 0.1 \text{ ms}$$

Final Answer: The one-hop transmission delay is **0.1 ms**.

Question 3

Consider a caravan of 10 cars (analogous to a 10-bit packet) passing through a toll booth (analogous to a link). The toll booth takes 12 seconds to service each car (analogous to bit transmission time), and the highway allows cars to propagate at a speed of 100 km/hr.

1. Calculate the total time to "push" the entire caravan through the first toll booth.
2. Calculate the propagation time for the last car to reach the second toll booth, 100 km away.

3. Determine the total time until the caravan is lined up before the second toll booth.

Answer

Step 1: Time to service entire caravan at first toll booth

$$t_{\text{service}} = \text{number of cars} \times \text{service time per car}$$

$$t_{\text{service}} = 10 \times 12 \text{ sec} = 120 \text{ sec}$$

Step 2: Propagation time from first to second toll booth

$$t_{\text{prop}} = \frac{\text{distance}}{\text{speed}}$$

$$t_{\text{prop}} = \frac{100 \text{ km}}{100 \text{ km/hr}} = 1 \text{ hr} = 60 \text{ min}$$

Step 3: Total time until caravan reaches second toll booth

$$t_{\text{total}} = t_{\text{service}} + t_{\text{prop}}$$

$$t_{\text{total}} = 120 \text{ sec} + 60 \text{ min} = 2 \text{ min} + 60 \text{ min} = 62 \text{ min}$$

Final Answer: The caravan will be lined up before the second toll booth after **62 minutes**.

Question 4

Suppose the cars now “propagate” at 1000 km/hr and the toll booth takes 1 minute to service each car.

1. Will any car arrive at the second toll booth before all cars are serviced at the first booth?
2. If so, calculate when the first car reaches the second booth and how many cars are still at the first booth at that time.

Answer

Step 1: Service time at first toll booth

$$t_{\text{service}} = 10 \times 1 \text{ min} = 10 \text{ min}$$

Step 2: Propagation time to second toll booth

$$t_{\text{prop}} = \frac{100 \text{ km}}{1000 \text{ km/hr}} = 0.1 \text{ hr} = 6 \text{ min}$$

Step 3: First car arrival at second booth - Leaves first booth after 1 min - Travels 6 min

$$t_{\text{arrival}} = 1 + 6 = 7 \text{ min}$$

Step 4: Cars still at first booth when first car arrives at second booth

$$\text{cars serviced by 7 min} = \frac{7 \text{ min}}{1 \text{ min/car}} = 7 \text{ cars}$$

$$\text{cars remaining at first booth} = 10 - 7 = 3 \text{ cars}$$

Conclusion: Yes, the first car arrives at the second toll booth after 7 minutes while 3 cars are still being serviced at the first booth.

Question 5

Suppose there is exactly one packet switch between a sending host and a receiving host. The transmission rates between the sending host and the switch and between the switch and the receiving host are R_1 and R_2 , respectively. Assuming that the switch uses store-and-forward packet switching, what is the total end-to-end delay to send a packet of length L ? (Ignore queuing, propagation delay, and processing delay.)

Answer

Step 1: Transmission from sending host to switch

$$t_1 = \frac{L}{R_1}$$

The sending host finishes transmitting the entire packet after L/R_1 seconds, and the switch receives it completely at this time.

Step 2: Transmission from switch to receiving host

$$t_2 = t_1 + \frac{L}{R_2}$$

Once the switch has the full packet, it begins transmission to the receiving host, taking L/R_2 seconds.

Step 3: Total end-to-end delay

$$d_{\text{end-to-end}} = \frac{L}{R_1} + \frac{L}{R_2}$$

Conclusion: The total end-to-end delay to send a packet of length L through one store-and-forward switch is:

$$d_{\text{total}} = \frac{L}{R_1} + \frac{L}{R_2}$$

Question 6

What advantage does a circuit-switched network have over a packet-switched network?
What advantages does TDM have over FDM in a circuit-switched network?

Answer

Step 1: Advantage of Circuit-Switched Network

- A **circuit-switched network** establishes a dedicated path between sender and receiver for the entire session.
- This ensures:
 - **Guaranteed bandwidth**
 - **Fixed and predictable delay**
 - **No congestion or queuing delay**

Advantage: Consistent and reliable end-to-end performance.

Step 2: Advantages of TDM over FDM in Circuit Switching

- **TDM (Time Division Multiplexing)**: Allocates the full channel bandwidth to each user for a short time slot.
- **FDM (Frequency Division Multiplexing)**: Divides total bandwidth into smaller frequency bands for each user.

TDM Advantages:

1. **Higher bandwidth efficiency** – No guard bands required.
2. **Better digital compatibility** – Ideal for modern digital systems.
3. **Lower interference** – Less crosstalk compared to FDM.

Conclusion: Circuit-switched networks offer guaranteed QoS through dedicated paths. Within such networks, **TDM** is preferred over **FDM** for its efficiency, digital alignment, and reduced interference.

Question 7

Suppose users share a 2 Mbps link. Also suppose each user transmits continuously at 1 Mbps when transmitting, but each user transmits only 20 percent of the time.

1. When circuit switching is used, how many users can be supported?
2. For the remainder of this problem, suppose packet switching is used. Why will there be essentially no queuing delay before the link if two or fewer users transmit

at the same time? Why will there be a queuing delay if three users transmit at the same time?

3. Find the probability that a given user is transmitting.
4. Suppose now there are three users. Find the probability that at any given time, all three users are transmitting simultaneously. Find the fraction of time during which the queue grows.

Answer

Step 1: Circuit switching capacity

Each user requires 1 Mbps when transmitting. The total link capacity is 2 Mbps.

$$\text{Maximum users (circuit switching)} = \frac{2 \text{ Mbps}}{1 \text{ Mbps/user}} = 2 \text{ users.}$$

2 users

Step 2: Packet switching — queuing explanation

- If at most two users transmit simultaneously, the total required bandwidth ≤ 2 Mbps, which matches the link capacity, so there is essentially no queuing delay. - If three users transmit simultaneously, total demand = 3 Mbps $>$ 2 Mbps, so the packet arrival rate exceeds the link capacity, leading to queuing delay.

Step 3: Probability that a given user is transmitting

Each user transmits 20% of the time:

$$P(\text{user transmitting}) = p = 0.2.$$

Step 4: Probability that all three users transmit simultaneously

Assuming each user transmits independently, the probability that all three transmit is given by the binomial probability formula:

$$P(k \text{ successes}) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Here, $n = 3$ (number of users), $k = 3$ (number transmitting), and $p = 0.2$. So,

$$P(\text{all 3 transmit}) = \binom{3}{3} p^3 (1-p)^{3-3}.$$

Simplifying:

$$\binom{3}{3} = 1, \quad (1-p)^0 = 1.$$

Therefore,

$$P(\text{all 3 transmit}) = 1 \cdot p^3 \cdot 1 = p^3.$$

Substituting $p = 0.2$,

$$P(\text{all 3 transmit}) = (0.2)^3 = 0.008.$$

Step 5: Fraction of time the queue grows

The queue grows only when all three users transmit simultaneously, i.e., when the total offered load exceeds the link capacity. Hence, the fraction of time the queue grows is:

Fraction of time queue grows = 0.008 (or 0.8% of the time).

Final Answers:

- (a) 2 users (circuit switching)
- (b) No queuing for ≤ 2 simultaneous transmitters; queuing occurs for 3
- (c) $P(\text{user transmitting}) = 0.2$
- (d) $P(\text{all 3 transmit}) = 0.008$; queue grows 0.8% of the time

Question 8

Why will two ISPs at the same level of the hierarchy often peer with each other? How does an IXP earn money?

Answer

Peering between ISPs: Two ISPs at the same level peer to exchange traffic directly without paying an upstream provider. This reduces cost, lowers latency, and improves performance when traffic volumes are similar.

IXP revenue: An Internet Exchange Point earns money by charging:

- Port or connection fees (based on bandwidth)
- Membership or access fees
- Colocation and facility service charges

Summary: ISPs peer to save cost and improve efficiency; IXPs earn by charging for connectivity and hosting services.

Question 9

Some content providers have created their own networks. Describe Google's network. What motivates content providers to create these networks?

Answer

Google's network: Google has built a private global network connecting its data centers, caching servers, and major ISPs. It uses high-capacity fiber links and places Google Global Cache (GGC) servers within ISP networks to deliver content closer to users.

Motivation: Content providers build their own networks to:

- Reduce dependency and costs of third-party ISPs.
- Improve performance by reducing latency and congestion.
- Gain better control over routing, reliability, and quality of service.

Summary: Private networks like Google's improve efficiency, lower cost, and enhance user experience by keeping traffic within controlled, high-speed paths.

Question 10

Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable?

Answer

Delay components: The end-to-end delay over a fixed route consists of:

1. **Processing delay** (d_{proc}) – Time to examine packet header and determine where to direct the packet.
2. **Queuing delay** (d_{queue}) – Time a packet waits in routing queues before transmission.
3. **Transmission delay** (d_{trans}) – Time to push all bits of the packet onto the link.
4. **Propagation delay** (d_{prop}) – Time for the signal to propagate through the medium.

Constant vs. variable:

- **Constant:** d_{proc} , d_{trans} , and d_{prop} (for fixed path and packet size).
- **Variable:** d_{queue} — depends on network congestion and traffic load.

Summary: End-to-end delay = $d_{proc} + d_{queue} + d_{trans} + d_{prop}$, with queuing delay being the only variable component.

Question 11

How long does it take a packet of length 1,000 bytes to propagate over a link of distance 2,500 km, propagation speed 2.5×10^8 m/s, and transmission rate 2 Mbps? More generally, how long does it take a packet of length L to propagate over a link of distance d , propagation speed s , and transmission rate R bps? Does this delay depend on packet length? Does this delay depend on transmission rate?

Answer

Given:

$$L = 1000 \text{ bytes} = 8000 \text{ bits}, \quad d = 2500 \text{ km} = 2.5 \times 10^6 \text{ m}, \\ s = 2.5 \times 10^8 \text{ m/s}, \quad R = 2 \text{ Mbps} = 2 \times 10^6 \text{ bps}.$$

Step 1: Propagation delay

$$d_{prop} = \frac{d}{s} = \frac{2.5 \times 10^6}{2.5 \times 10^8} = 0.01 \text{ s} = 10 \text{ ms}.$$

Step 2: Transmission delay

$$d_{trans} = \frac{L}{R} = \frac{8000}{2 \times 10^6} = 0.004 \text{ s} = 4 \text{ ms}.$$

Step 3: Total delay

$$d_{total} = d_{prop} + d_{trans} = 10 \text{ ms} + 4 \text{ ms} = 14 \text{ ms}.$$

General formula:

$$d_{total} = \frac{d}{s} + \frac{L}{R}.$$

Dependence:

- **Propagation delay** (d/s) — depends only on distance d and propagation speed s , *not* on packet length L or rate R .
- **Transmission delay** (L/R) — depends on packet length and transmission rate.

Summary: Total delay = 10 ms (propagation) + 4 ms (transmission) = 14 ms. Propagation delay does *not* depend on packet length or rate; transmission delay does.

Question 12

Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates $R_1 = 500$ kbps, $R_2 = 2$ Mbps, and $R_3 = 1$ Mbps.

- (a) Assuming no other traffic in the network, what is the throughput for the file transfer?

- (b) Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?
- (c) Repeat (a) and (b), but now with R_2 reduced to 100 kbps.

Answer

Given:

$$R_1 = 500 \text{ kbps}, \quad R_2 = 2 \text{ Mbps}, \quad R_3 = 1 \text{ Mbps},$$

$$\text{File size} = 4 \text{ MB} = 4 \times 10^6 \text{ bytes} = 32 \times 10^6 \text{ bits}.$$

(a) Throughput (no congestion): Throughput is determined by the **bottleneck link**, i.e., the link with the lowest transmission rate.

$$\text{Throughput} = \min(R_1, R_2, R_3) = \min(500, 2000, 1000) \text{ kbps} = 500 \text{ kbps}.$$

$$\boxed{\text{Throughput} = 500 \text{ kbps}}$$

(b) Transfer time:

$$\text{Time} = \frac{\text{File size}}{\text{Throughput}} = \frac{32 \times 10^6 \text{ bits}}{500 \times 10^3 \text{ bps}} = 64 \text{ s}.$$

$$\boxed{\text{Transfer time} = 64 \text{ seconds.}}$$

(c) With $R_2 = 100 \text{ kbps}$:

$$\text{Throughput} = \min(500, 100, 1000) \text{ kbps} = 100 \text{ kbps}.$$

$$\text{Time} = \frac{32 \times 10^6}{100 \times 10^3} = 320 \text{ s}.$$

$$\boxed{\text{Throughput} = 100 \text{ kbps}, \quad \text{Transfer time} = 320 \text{ seconds.}}$$

Summary:

Case	Throughput	Transfer Time
$R_2 = 2 \text{ Mbps}$	500 kbps	64 s
$R_2 = 100 \text{ kbps}$	100 kbps	320 s

Question 13

Suppose end system A wants to send a large file to end system B. At a very high level, describe how end system A creates packets from the file. When one of these packets arrives at a router, what information in the packet does the router use to determine the link onto which the packet is forwarded? Why is packet switching in the Internet

analogous to driving from one city to another and asking directions along the way?

Answer

File to packets: End system A divides the large file into smaller, fixed-size chunks called **packets**. Each packet contains:

- A **header** — includes source and destination IP addresses.
- A **payload** — a portion of the original file's data.

Router forwarding decision: When a packet arrives at a router, the router:

- Examines the **destination IP address** in the header.
- Uses its **forwarding table (routing table)** to determine the appropriate outgoing link.

Analogy — asking directions: Packet switching is like driving from one city to another and asking for directions at each intersection — each router (like a city junction) makes an independent decision on the next path based on the packet's destination.

Summary: Packets are formed by segmenting the file; routers forward them using destination IP addresses; and packet switching is dynamic and hop-by-hop, similar to following directions along the route.

Question 14

What is an application-layer message? A transport-layer segment? A network-layer datagram? A link-layer frame?

Answer

Application-layer message:

- The data created by an application process (e.g., an email message, HTTP request, or file data).
- Represents the original information to be transmitted over the network.

Transport-layer segment:

- The transport layer (TCP or UDP) encapsulates the application message.
- It adds a **transport header** containing fields such as source/destination port numbers, sequence numbers, and error checking.
- The result is a **segment**.

Network-layer datagram:

- The network layer (IP) encapsulates the transport segment.
- It adds a **network header** with source and destination IP addresses.
- The resulting unit is called a **datagram**.

Link-layer frame:

- The link layer (Ethernet, Wi-Fi, etc.) encapsulates the datagram.
- It adds a **link-layer header and trailer** containing physical (MAC) addresses and error-detection bits.
- The complete unit is a **frame**.

Summary:

Application Message → Transport Segment → Network Datagram → Link Frame

Each layer adds its own header (and possibly trailer) before passing data to the next lower layer.

Question 15

Which layers in the Internet protocol stack does a router process? Which layers does a link-layer switch process? Which layers does a host process?

Answer

Router:

- Processes up to the **network layer**.
- Examines the **IP header** to determine how to forward packets.
- Operates at the **physical, link, and network** layers.

Link-layer switch:

- Operates up to the **link layer** only.
- Uses **MAC addresses** to forward frames within a local network (LAN).
- Does *not* process IP or transport-layer headers.

Host:

- Processes all layers — from the **physical layer** up to the **application layer**.
- Acts as both the source and destination for data.
- Handles **application**, **transport**, **network**, **link**, and **physical** layer functions.

Summary Table:

Device	Layers Processed
Router	Physical, Link, Network
Link-layer Switch	Physical, Link
Host	All Layers (Physical → Application)

Question 16

What is self-replicating malware?

Answer

Self-replicating malware is a type of malicious software that can make copies of itself and spread to other systems automatically, without user intervention. It can replicate through networks, email attachments, or infected files.

Examples:

- **Virus** – Attaches itself to legitimate programs and spreads when those programs are executed.
- **Worm** – Spreads independently over networks without needing to attach to other programs.

Key characteristic: The ability to *replicate and propagate autonomously*, making it highly effective at spreading rapidly.

Question 17

Describe how a botnet can be created and how it can be used for a DDoS attack.

Answer

Botnet creation (brief):

- **Infection:** Malware (phishing, drive-by, unpatched services) infects devices.
- **Enroll:** Infected hosts contact a command-and-control (C&C) to receive instructions.
- **Maintain:** Attacker grows and maintains the bot pool (updates, persistence, P2P)

or resilient C&C).

Use in DDoS (brief):

- Attacker commands many bots to send coordinated traffic to a target.
- Types: volumetric floods (saturate bandwidth), protocol/state exhaustion, application-layer floods, or reflection/amplification via spoofing.
- Result: aggregated traffic overwhelms the target, causing service disruption.

Quick mitigations: patching, endpoint protection, traffic filtering/scrubbing, rate limiting, and ISP upstream filtering.

Question 18

Design and describe an application-level protocol to be used between an automatic teller machine (ATM) and a bank's centralized computer. Your protocol should allow:

- Card and password verification
- Account balance query
- Account withdrawal

It should handle insufficient funds and assume a reliable end-to-end transport service.

Answer

Protocol name: ATM-BANK Secure Transaction Protocol (ABSTP)

Assumption: Uses a reliable, connection-oriented transport (e.g., TCP). Messages are delivered in order and without loss.

Message Types:

1. AUTH_REQ(card_id, password) — ATM → Bank: Request authentication
2. AUTH_RESP(status) — Bank → ATM: "OK" or "FAIL"
3. BAL_REQ(card_id) — ATM → Bank: Request account balance
4. BAL_RESP(balance) — Bank → ATM: Send current balance
5. WITHDRAW_REQ(card_id, amount) — ATM → Bank: Request withdrawal
6. WITHDRAW_RESP(status, new_balance) — Bank → ATM: Confirm or deny (e.g., "INSUFFICIENT_FUNDS")

ATM Actions:

- Send authentication message; if failed, terminate session.
- On success, allow balance query or withdrawal.
- Display responses to user and dispense cash if authorized.

Bank Actions:

- Verify credentials; send status.
- On balance request, look up account and return balance.
- On withdrawal request, check funds:
 - If sufficient — deduct amount and return updated balance.
 - If insufficient — send “INSUFFICIENT_FUNDS”.

Normal operation (no errors):

ATM $\longrightarrow$ Bank: AUTH_REQ(card, pass)
 ATM $\longleftarrow$ Bank: AUTH_RESP(OK)
 ATM $\longrightarrow$ Bank: WITHDRAW_REQ(card, \$100)
 ATM $\longleftarrow$ Bank: WITHDRAW_RESP(OK, new_balance)
 ATM dispenses cash.

Summary: ABSTP ensures secure and reliable interaction between ATM and bank server using structured message exchanges and assumes reliable transport.

Question 19

Equation gives the end-to-end delay for sending one packet of length L over N links, each of transmission rate R :

$$d_{end-to-end} = N \frac{L}{R}$$

Now, generalize this formula for the case of sending P packets back-to-back over the N links.

Answer

Explanation: In store-and-forward packet switching, each router must receive the *entire packet* before forwarding it to the next link.

If instead routers forwarded bits as soon as they arrived (cut-through switching), the total delay would be only $\frac{L}{R}$, since bits would flow continuously. However, in real networks, routers must fully receive, store, and process packets before sending them onward.

Derivation for P packets:

- At time $\frac{L}{R}$, the first packet reaches the first router, and the source starts sending

the second packet.

- At time $2\frac{L}{R}$, the first packet reaches the destination, and the router holds the second packet.
- Following this pattern, each new packet arrives one $\frac{L}{R}$ later.

Thus, when the last (i.e., P^{th}) packet reaches the destination:

$$T_P = N\frac{L}{R} + (P - 1)\frac{L}{R} = (N + P - 1)\frac{L}{R}$$

Final formula:

$$T_P = (N + P - 1)\frac{L}{R}$$

Interpretation: - $N\frac{L}{R}$ = time for the first packet to reach the destination. - $(P - 1)\frac{L}{R}$ = additional time for the remaining packets to arrive. - Total delay increases linearly with both the number of links N and packets P .

Question 20

Consider an application that transmits data at a steady rate (for example, the sender generates an N -bit unit of data every k time units, where k is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answer:

- a) Would a packet-switched network or a circuit-switched network be more appropriate for this application? Why?
- b) Suppose that a packet-switched network is used and the only traffic in this network comes from such applications as described above. Furthermore, assume that the sum of the application data rates is less than the capacities of each and every link. Is some form of congestion control needed? Why?

Answer

(a) Appropriate network type: A **circuit-switched network** is more suitable for this application because:

- The data transmission rate is steady and predictable.
- Bandwidth can be reserved for the entire session, ensuring consistent performance.
- The setup overhead is minimal relative to the long session duration.

(b) Congestion control in packet switching: If a packet-switched network is used and the total traffic rate is less than the capacity of every link:

- There will be **no congestion** since link capacity is sufficient to handle all traffic.
- Hence, **no congestion control** mechanism is required.
- Queuing delay will remain minimal even if all applications transmit simultaneously.

Conclusion: Circuit switching best fits long-lived, steady-rate flows. Packet switching can handle such flows efficiently without congestion control, provided link capacities exceed aggregate traffic rates.

Question 21

Consider the circuit-switched network in Figure 1. There are four circuits on each link. Label the four switches *A*, *B*, *C*, and *D*, going in the clockwise direction.

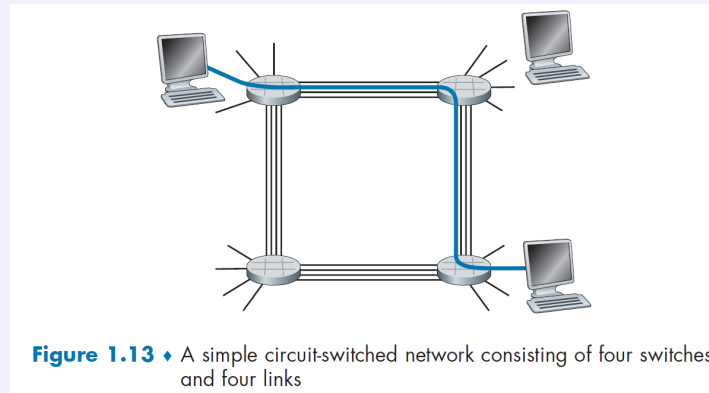


Figure 1: Circuit-switched network with four switches and four circuits per link.

- What is the maximum number of simultaneous connections that can be in progress at any one time in this network?
- Suppose that all connections are between switches *A* and *C*. What is the maximum number of simultaneous connections that can be in progress?
- Suppose we want to make four connections between switches *A* and *C*, and another four connections between switches *B* and *D*. Can we route these calls through the four links to accommodate all eight connections?

Answer

- Each link can carry 4 connections. There are 4 links connecting adjacent switches (*A*–*B*, *B*–*C*, *C*–*D*, *D*–*A*). Therefore, the maximum number of simultaneous connections in the network is:

$$4 \text{ connections/link} \times 4 \text{ links} = 16 \text{ connections.}$$

- For connections only between *A* and *C*, the paths must pass through either *B* or

D. Each path can carry 4 connections, giving a total of:

$$4 \text{ connections via B} + 4 \text{ connections via D} = 8 \text{ connections.}$$

c) Yes, it is possible. We can route the connections as follows:

- For *A–C*: 2 connections via *B* and 2 connections via *D*.
- For *B–D*: 2 connections via *A* and 2 connections via *C*.

This ensures that no link carries more than 4 connections, so all 8 connections can be accommodated.

Question 22

Review the car-caravan analogy in Question 3. Assume a propagation speed of 100 km/hr.

1. Suppose the caravan travels 175 km, beginning in front of one toll booth, passing through a second toll booth, and finishing just after a third toll booth. Calculate the end-to-end delay.
2. Repeat part (a), now assuming that there are eight cars in the caravan instead of ten.

Answer

Given: - Propagation speed: $v = 100$ km/hr

- Distance traveled: $d = 175$ km

- Service time per car at each toll booth: 12 sec

- Number of toll booths: 3

Step 1: Propagation time for 175 km

$$t_{\text{prop}} = \frac{d}{v} = \frac{175 \text{ km}}{100 \text{ km/hr}} = 1.75 \text{ hr} = 105 \text{ min}$$

Step 2: Total service time at toll booths

$$t_{\text{service}} = \text{number of cars} \times \text{service time per car} \times \text{number of toll booths}$$

(a) 10-car caravan:

$$t_{\text{service}} = 10 \times 12 \text{ sec} \times 3 = 360 \text{ sec} = 6 \text{ min}$$

Step 3: End-to-end delay

$$t_{\text{total}} = t_{\text{prop}} + t_{\text{service}} = 105 \text{ min} + 6 \text{ min} = 111 \text{ min}$$

(b) **8-car caravan:**

$$t_{\text{service}} = 8 \times 12 \text{ sec} \times 3 = 288 \text{ sec} = 4.8 \text{ min} \approx 4.8 \text{ min}$$

$$t_{\text{total}} = 105 \text{ min} + 4.8 \text{ min} \approx 109.8 \text{ min} \approx 110 \text{ min}$$

Final Answers:

- (a) End-to-end delay for 10-car caravan: **111 minutes**
- (b) End-to-end delay for 8-car caravan: **110 minutes**

Question 23

This elementary problem explores propagation delay and transmission delay. Consider two hosts, A and B, connected by a single link of rate R bps. The hosts are separated by m meters, and the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.

- Express the propagation delay, d_{prop} , in terms of m and s .
- Determine the transmission time of the packet, d_{trans} , in terms of L and R .
- Ignoring processing and queuing delays, obtain an expression for the end-to-end delay.
- Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{\text{trans}}$, where is the last bit of the packet?
- Suppose $d_{\text{prop}} > d_{\text{trans}}$. At time $t = d_{\text{trans}}$, where is the first bit of the packet?
- Suppose $d_{\text{prop}} < d_{\text{trans}}$. At time $t = d_{\text{trans}}$, where is the first bit of the packet?
- Suppose $s = 2.5 \times 10^8$ m/s, $L = 1500$ bytes, and $R = 10$ Mbps. Find the distance m so that d_{prop} equals d_{trans} .

Answer

a) **Propagation delay:**

$$d_{\text{prop}} = \frac{m}{s}$$

b) **Transmission delay:**

$$d_{\text{trans}} = \frac{L}{R}$$

c) **End-to-end delay (ignoring processing and queuing):**

$$d_{\text{end-to-end}} = d_{\text{trans}} + d_{\text{prop}} = \frac{L}{R} + \frac{m}{s}$$

- d) **Location of last bit at $t = d_{\text{trans}}$:** At $t = d_{\text{trans}}$, the last bit has just left Host A and is starting to propagate along the link. The first bit is already in transit toward Host B.
- e) **If $d_{\text{prop}} > d_{\text{trans}}$:** At $t = d_{\text{trans}}$, the first bit is still on the link, not yet arrived at Host B.
- f) **If $d_{\text{prop}} < d_{\text{trans}}$:** At $t = d_{\text{trans}}$, the first bit has already arrived at Host B (since it propagated faster than the time to transmit the entire packet).
- g) **Distance m such that $d_{\text{prop}} = d_{\text{trans}}$:**

$$d_{\text{prop}} = d_{\text{trans}} \implies \frac{m}{s} = \frac{L}{R} \implies m = \frac{L}{R} \cdot s$$

Convert $L = 1500$ bytes to bits: $L = 1500 \times 8 = 12000$ bits. $R = 10$ Mbps $= 10 \times 10^6$ bps, $s = 2.5 \times 10^8$ m/s.

$$m = \frac{12000}{10 \times 10^6} \times 2.5 \times 10^8$$

$$m = 0.0012 \times 2.5 \times 10^8 = 3 \times 10^5 \text{ m} = 300 \text{ km}$$

Answer: $m = 300$ km

Question 24

Consider sending real-time voice from Host A to Host B over a packet-switched network (VoIP).

- Host A converts analog voice to a digital 64 kbps bit stream on the fly.
- Host A groups the bits into 56-byte packets.
- There is a single link between A and B with transmission rate $R = 10$ Mbps and propagation delay $d_{\text{prop}} = 10$ msec.
- As soon as Host A gathers a packet, it sends it to Host B.
- Host B converts the packet's bits to analog signal as soon as it receives the entire packet.

Question: How much time elapses from the creation of a bit at Host A until it is decoded at Host B?

Answer

Step 1: Calculate packet size in bits

$$\text{Packet size } L = 56 \text{ bytes} \times 8 \text{ bits/byte} = 448 \text{ bits}$$

Step 2: Calculate packetization delay This is the time to accumulate bits into a

packet:

$$d_{\text{packetization}} = \frac{L}{\text{voice bit rate}} = \frac{448 \text{ bits}}{64 \times 10^3 \text{ bps}}$$

$$d_{\text{packetization}} = 0.007 \text{ sec} = 7 \text{ ms}$$

Step 3: Transmission delay Time to push the packet onto the link:

$$d_{\text{trans}} = \frac{L}{R} = \frac{448 \text{ bits}}{10 \times 10^6 \text{ bps}}$$

$$d_{\text{trans}} \approx 0.0000448 \text{ sec} \approx 0.045 \text{ ms}$$

Step 4: Propagation delay Given:

$$d_{\text{prop}} = 10 \text{ ms}$$

Step 5: Total end-to-end delay

$$d_{\text{total}} = d_{\text{packetization}} + d_{\text{trans}} + d_{\text{prop}}$$

$$d_{\text{total}} \approx 7 \text{ ms} + 0.045 \text{ ms} + 10 \text{ ms} \approx 17.045 \text{ ms}$$

Final Answer: The elapsed time from creation of a bit at Host A until it is decoded at Host B is approximately

17 ms

Question 25

Suppose users share a 10 Mbps link. Each user requires 200 kbps when transmitting, but each user transmits only 10% of the time.

- a) When circuit switching is used, how many users can be supported?
- b) For packet switching, find the probability that a given user is transmitting.
- c) Suppose there are 120 users. Find the probability that at any given time, exactly n users are transmitting simultaneously. (Hint: Use the binomial distribution.)
- d) Find the probability that 51 or more users are transmitting simultaneously.

Answer

Step a: Circuit switching capacity

Each user requires 200 kbps. Total link capacity is 10 Mbps:

$$\text{Max users (circuit)} = \frac{10 \text{ Mbps}}{0.2 \text{ Mbps/user}} = 50 \text{ users}$$

50 users

Step b: Probability that a given user is transmitting

Given that each user transmits 10% of the time:

$$P(\text{user transmitting}) = 0.1$$

$$\boxed{0.1}$$

Step c: Probability exactly n users are transmitting (binomial distribution)

Let $N = 120$ users and $p = 0.1$. The probability that exactly n users transmit is:

$$P(X = n) = \binom{N}{n} p^n (1 - p)^{N-n} = \binom{120}{n} (0.1)^n (0.9)^{120-n}$$

Step d: Probability that 51 or more users transmit simultaneously

$$P(X \geq 51) = \sum_{n=51}^{120} \binom{120}{n} (0.1)^n (0.9)^{120-n}$$

Note: This probability is extremely small because the expected number of users transmitting simultaneously is

$$E[X] = Np = 120 \times 0.1 = 12$$

and 51 is far above the mean. In practice, $P(X \geq 51) \approx 0$.

Question 26

Consider the discussion of packet switching versus circuit switching. In the original example, a 1 Mbps link carried users generating 100 kbps when busy, each busy with probability $p = 0.1$. Now, suppose the link is replaced with a 1 Gbps link.

- What is N , the maximum number of users that can be supported simultaneously under circuit switching?
- For packet switching with a population of M users, give a formula (in terms of p , M , N) for the probability that more than N users are sending data simultaneously.

Answer**Step a: Maximum users under circuit switching**

Each user requires 100 kbps when transmitting. Link capacity is 1 Gbps:

$$N = \frac{\text{link capacity}}{\text{user rate}} = \frac{1 \text{ Gbps}}{100 \text{ kbps}}$$

$$N = \frac{10^9}{10^5} = 10,000 \text{ users}$$

$$N = 10,000$$

Step b: Probability more than N users transmit simultaneously under packet switching

Let M be the total number of users, and each user is transmitting with probability p . Define a random variable X as the number of users transmitting simultaneously. Then $X \sim \text{Binomial}(M, p)$.

The probability that more than N users transmit is:

$$P(X > N) = \sum_{k=N+1}^M \binom{M}{k} p^k (1-p)^{M-k}$$

This formula expresses the probability of overload in a packet-switched network given M users, transmission probability p , and circuit-switched capacity N .

Question 27

Consider a packet of length L that begins at end system A and travels over three links to a destination. These links are connected by two packet switches. Let d_i , s_i , and R_i denote the length, propagation speed, and transmission rate of link i , for $i = 1, 2, 3$. Each packet switch delays a packet by d_{proc} . Assume no queuing delays.

- Express the total end-to-end delay in terms of d_i , s_i , R_i , L , and d_{proc} .
- Suppose $L = 1,500$ bytes, $s_i = 2.5 \times 10^8$ m/s, $R_i = 2.5$ Mbps, $d_{\text{proc}} = 3$ ms, $d_1 = 5,000$ km, $d_2 = 4,000$ km, $d_3 = 1,000$ km. Compute the total end-to-end delay.

Answer

Step a: General expression for end-to-end delay

For each link i , the transmission delay is:

$$d_{\text{trans},i} = \frac{L}{R_i}$$

and the propagation delay is:

$$d_{\text{prop},i} = \frac{d_i}{s_i}$$

The total end-to-end delay over three links and two packet switches is:

$$d_{\text{end-to-end}} = \sum_{i=1}^3 (d_{\text{trans},i} + d_{\text{prop},i}) + 2 d_{\text{proc}}$$

$$d_{\text{end-to-end}} = \sum_{i=1}^3 \frac{L}{R_i} + \sum_{i=1}^3 \frac{d_i}{s_i} + 2d_{\text{proc}}$$

Step b: Numerical computation

Given values:

$$L = 1,500 \text{ bytes} = 12,000 \text{ bits}, \quad R_i = 2.5 \text{ Mbps} = 2.5 \times 10^6 \text{ bps}$$

$$d_1 = 5,000 \text{ km} = 5 \times 10^6 \text{ m}, \quad d_2 = 4,000 \text{ km} = 4 \times 10^6 \text{ m}, \quad d_3 = 1,000 \text{ km} = 1 \times 10^6 \text{ m}$$

$$s_i = 2.5 \times 10^8 \text{ m/s}, \quad d_{\text{proc}} = 3 \text{ ms} = 0.003 \text{ s}$$

Transmission delays:

$$d_{\text{trans},i} = \frac{12,000}{2.5 \times 10^6} \approx 0.0048 \text{ s} = 4.8 \text{ ms per link}$$

Total for three links:

$$3 \times 4.8 \text{ ms} = 14.4 \text{ ms}$$

Propagation delays:

$$d_{\text{prop},1} = \frac{5 \times 10^6}{2.5 \times 10^8} = 0.02 \text{ s} = 20 \text{ ms}$$

$$d_{\text{prop},2} = \frac{4 \times 10^6}{2.5 \times 10^8} = 0.016 \text{ s} = 16 \text{ ms}$$

$$d_{\text{prop},3} = \frac{1 \times 10^6}{2.5 \times 10^8} = 0.004 \text{ s} = 4 \text{ ms}$$

Total propagation delay: $20 + 16 + 4 = 40 \text{ ms}$

Processing delays:

$$2 \times d_{\text{proc}} = 2 \times 3 \text{ ms} = 6 \text{ ms}$$

Total end-to-end delay:

$$d_{\text{end-to-end}} = 14.4 + 40 + 6 \approx 60.4 \text{ ms}$$

Final Answer:

$d_{\text{end-to-end}} \approx 60.4 \text{ ms}$

Question 28

Using the previous problem's data:

$$L = 1,500 \text{ bytes} = 12,000 \text{ bits}, \quad R_1 = R_2 = R_3 = 2.5 \text{ Mbps},$$

$$d_1 = 5,000 \text{ km}, \quad d_2 = 4,000 \text{ km}, \quad d_3 = 1,000 \text{ km}, \quad s_i = 2.5 \times 10^8 \text{ m/s},$$

$$d_{\text{proc}} = 0$$

Assuming cut-through switching (bits transmitted immediately, no store-and-forward), compute the end-to-end delay.

Answer

Step 1: Transmission delay of the first link

$$d_{\text{trans}} = \frac{L}{R} = \frac{12,000 \text{ bits}}{2.5 \times 10^6 \text{ bps}} \approx 0.0048 \text{ s} = 4.8 \text{ ms}$$

Step 2: Propagation delays for all three links

$$d_{\text{prop},1} = \frac{5 \times 10^6}{2.5 \times 10^8} = 0.02 \text{ s} = 20 \text{ ms}$$

$$d_{\text{prop},2} = \frac{4 \times 10^6}{2.5 \times 10^8} = 0.016 \text{ s} = 16 \text{ ms}$$

$$d_{\text{prop},3} = \frac{1 \times 10^6}{2.5 \times 10^8} = 0.004 \text{ s} = 4 \text{ ms}$$

Step 3: Total end-to-end delay

$$d_{\text{end-to-end}} = d_{\text{trans}} + \sum_{i=1}^3 d_{\text{prop},i} = 4.8 + 20 + 16 + 4 \text{ ms}$$
$$d_{\text{end-to-end}} = 44.8 \text{ ms}$$

Final Answer:

$$d_{\text{end-to-end}} \approx 44.8 \text{ ms}$$

Note: Compared to the store-and-forward case (60.4 ms), cut-through switching reduces the delay by about 15.6 ms.

Question 29

A packet switch receives a packet and must forward it on an outbound link. When the packet arrives:

- One packet is already being transmitted (halfway done)
- Four other packets are waiting in the queue

All packets are 1,500 bytes, and the link rate is 2.5 Mbps.

1. What is the queuing delay for the arriving packet?
2. More generally, if all packets have length L , the transmission rate is R , x bits of the currently-transmitting packet have been transmitted, and n packets are already in the queue, what is the queuing delay?

Answer

Step 1: Transmission time of one packet

Packet length $L = 1,500$ bytes = 12,000 bits, link rate $R = 2.5$ Mbps = 2.5×10^6 bps:

$$d_{\text{trans}} = \frac{L}{R} = \frac{12,000}{2.5 \times 10^6} \approx 0.0048 \text{ s} = 4.8 \text{ ms}$$

Step 2: Remaining transmission of the in-progress packet

Half of the first packet has already been transmitted:

$$d_{\text{remaining}} = \frac{L/2}{R} = \frac{6,000}{2.5 \times 10^6} \approx 0.0024 \text{ s} = 2.4 \text{ ms}$$

Step 3: Queuing delay for the 4 packets in the queue

$$d_{\text{queue}} = 4 \times d_{\text{trans}} = 4 \times 4.8 \text{ ms} = 19.2 \text{ ms}$$

Step 4: Total queuing delay for the arriving packet

$$D_{\text{queue}} = d_{\text{remaining}} + d_{\text{queue}} = 2.4 + 19.2 = 21.6 \text{ ms}$$

Step 5: General formula

Let each packet have length L , transmission rate R , x bits of the currently transmitting packet already sent, and n packets already in the queue. Then:

$$D_{\text{queue}} = \frac{L - x}{R} + n \frac{L}{R} = \frac{(n + 1)L - x}{R}$$

Answer:

$$D_{\text{queue}} = 21.6 \text{ ms (numerical)},$$

$$D_{\text{queue}} = \frac{(n + 1)L - x}{R} \text{ (general)}$$

Question 30

(a) Suppose N packets arrive simultaneously to a link at which no packets are currently being transmitted or queued. Each packet has length L , and the link has transmission rate R .

(b) Now suppose N such packets arrive to the link every LN/R seconds. What is the average queuing delay of a packet in this case?

Answer

Part (a): Packets arrive simultaneously

- Let each packet have transmission time:

$$d_{\text{trans}} = \frac{L}{R}$$

- Packets are transmitted in order of arrival. The first packet experiences **no queuing delay**, the second waits for the first to transmit, the third waits for the first two, etc.
- Queuing delays for each packet:

$$0, d_{\text{trans}}, 2d_{\text{trans}}, \dots, (N-1)d_{\text{trans}}$$

- Average queuing delay:

$$\bar{D}_{\text{queue}} = \frac{0 + d_{\text{trans}} + 2d_{\text{trans}} + \dots + (N-1)d_{\text{trans}}}{N} = \frac{d_{\text{trans}}(0 + 1 + 2 + \dots + (N-1))}{N}$$

- Sum of first $N-1$ integers:

$$0 + 1 + \dots + (N-1) = \frac{(N-1)N}{2}$$

- Therefore:

$$\bar{D}_{\text{queue}} = \frac{d_{\text{trans}} \cdot \frac{N(N-1)}{2}}{N} = \frac{(N-1)L}{2R}$$

Part (b): Packets arrive every LN/R seconds

- In this case, the link has just enough time to transmit all N packets before the next batch arrives.
- No packet in the new batch sees queuing due to the previous batch because:

$$\text{arrival interval} = \frac{LN}{R} = Nd_{\text{trans}}$$

- The first packet of each batch sees **no queuing**, the second waits d_{trans} , ..., the last waits $(N-1)d_{\text{trans}}$.
- Therefore, the **average queuing delay per packet** is the same as in part (a):

$$\bar{D}_{\text{queue}} = \frac{(N-1)L}{2R}$$

Answer:

$$\boxed{\bar{D}_{\text{queue}} = \frac{(N-1)L}{2R}}$$

for both cases.

Question 31

Consider the queuing delay in a router buffer. Let I denote the traffic intensity, that is, $I = \frac{La}{R}$. Suppose that the queuing delay takes the form:

$$d_{\text{queue}} = \frac{IL}{R(1-I)}, \quad I \leq 1$$

1. Provide a formula for the total delay, that is, the queuing delay plus the transmission delay.
2. Plot the total delay as a function of L/R .

Answer

Step 1: Transmission delay

$$d_{\text{trans}} = \frac{L}{R}$$

Step 2: Total delay

Total delay is the sum of transmission and queuing delays:

$$d_{\text{total}} = d_{\text{trans}} + d_{\text{queue}} = \frac{L}{R} + \frac{IL}{R(1-I)}$$

Substitute $I = \frac{La}{R}$:

$$d_{\text{total}} = \frac{L}{R} + \frac{\frac{La}{R} \cdot L}{R(1 - \frac{La}{R})} = \frac{L}{R} + \frac{L^2a/R}{R - La} = \frac{L}{R} + \frac{L^2a}{R(R - La)}$$

$$d_{\text{total}} = \frac{L}{R} + \frac{L^2a}{R(R - La)}$$

Step 3: Plotting the total delay as a function of L/R

Let $x = L/R$. Then $I = ax$, and the total delay becomes:

$$d_{\text{total}} = x + \frac{ax^2}{1 - ax}, \quad ax \leq 1$$

This is a convex function that increases slowly for small x and rises sharply as x approaches $1/a$.

Question 32

The total number of packets in a system includes those in the buffer and the packet that is currently being transmitted. Using Little's formula:

$$N = a \cdot d$$

where N is the total number of packets, a is the average arrival rate, and d is the total delay (queuing plus transmission), compute a given the following:

- There are 20 packets in the buffer on average.
- One packet is being transmitted.

- The queuing delay is 10 msec.
- The link's transmission rate is 100 packets/sec.

Answer

Step 1: Total number of packets in the system

$$N = \text{packets in buffer} + \text{packet being transmitted} = 20 + 1 = 21$$

Step 2: Transmission delay per packet

$$d_{\text{trans}} = \frac{1}{\text{link rate}} = \frac{1}{100} \text{ sec} = 0.01 \text{ sec}$$

Step 3: Total delay per packet

$$d = d_{\text{queue}} + d_{\text{trans}} = 0.01 + 0.01 = 0.02 \text{ sec}$$

Step 4: Apply Little's formula

$$N = a \cdot d \implies a = \frac{N}{d} = \frac{21}{0.02} = 1050 \text{ packets/sec}$$

Answer:

$$a = 1050 \text{ packets/sec}$$

Question 33

Suppose two hosts, A and B, are separated by 20,000 kilometers and are connected by a direct link of $R = 5$ Mbps. The propagation speed over the link is $s = 2.5 \times 10^8$ m/s.

1. Calculate the bandwidth-delay product, $R \cdot d_{\text{prop}}$.
2. Consider sending a file of 800,000 bits from Host A to Host B as one large message. What is the maximum number of bits that will be in the link at any given time?
3. Provide an interpretation of the bandwidth-delay product.
4. What is the width (in meters) of a bit in the link? Is it longer than a football field?
5. Derive a general expression for the width of a bit in terms of propagation speed s , transmission rate R , and link length m .

Answer

Given:

$$m = 20,000 \text{ km} = 2 \times 10^7 \text{ m}, \quad R = 5 \text{ Mbps} = 5 \times 10^6 \text{ bps}, \quad s = 2.5 \times 10^8 \text{ m/s}$$

(a) Bandwidth-delay product The propagation delay is:

$$d_{\text{prop}} = \frac{m}{s} = \frac{2 \times 10^7}{2.5 \times 10^8} = 0.08 \text{ sec}$$

Bandwidth-delay product:

$$R \cdot d_{\text{prop}} = 5 \times 10^6 \cdot 0.08 = 400,000 \text{ bits}$$

(b) Maximum number of bits in the link If the file size is 800,000 bits, the maximum number of bits in the link is limited by the smaller of the file size and the bandwidth-delay product. Since the file is larger than the bandwidth-delay product:

$$\text{Max bits in link} = 400,000 \text{ bits}$$

(c) Interpretation of bandwidth-delay product The bandwidth-delay product represents the number of bits that can fill the link: the maximum amount of data “in flight” at any instant. It determines the link’s capacity to hold bits during transmission.

(d) Width of a bit in meters

$$\text{Bit length} = \frac{\text{link length}}{\text{number of bits in link}} = \frac{2 \times 10^7 \text{ m}}{400,000} = 50 \text{ m/bit}$$

A standard football field is approximately 100 meters, so each bit is about half a football field in length.

(e) General expression for bit width Let w denote the width of a bit:

$$w = \frac{\text{link length}}{\text{number of bits in link}} = \frac{m}{R \cdot d_{\text{prop}}} = \frac{m}{R \cdot (m/s)} = \frac{s}{R}$$

So the width of a bit depends only on the propagation speed s and transmission rate R , independent of the link length m .

Question 34

Suppose you would like to urgently deliver 50 terabytes of data from Boston to Los Angeles. You have available a 100 Mbps dedicated link for data transfer. Would you prefer to transmit the data via this link or instead use FedEx overnight delivery? Explain.

Answer

Data size: 50 terabytes = $50 \times 10^{12} \times 8 = 4 \times 10^{14}$ bits.

Link speed: $R = 100 \text{ Mbps} = 10^8 \text{ bps}$.

Time to transmit:

$$t = \frac{\text{Data size}}{R} = \frac{4 \times 10^{14}}{10^8} = 4 \times 10^6 \text{ seconds} \approx 46 \text{ days}$$

Conclusion: Using the 100 Mbps link would take 46 days, which is far longer than FedEx overnight delivery (1 day). Therefore, FedEx is preferred.

Question 35

Referring to problem Q34, suppose we can modify R . For what value of R is the width of a bit as long as the length of the link?

Answer

Let the link length be $m = 4,500 \text{ km} \approx 4.5 \times 10^6 \text{ m}$, and propagation speed $s \approx 3 \times 10^8 \text{ m/s}$.

Bit width formula:

$$\text{Bit width} = \frac{s}{R}$$

For bit width = link length m :

$$R = \frac{s}{m} = \frac{3 \times 10^8}{4.5 \times 10^6} \approx 66.7 \text{ bps}$$

Conclusion: If $R \approx 66.7 \text{ bps}$, each bit is physically as long as the link.

Question 36

Consider problem Q34 but now with a link of $R = 500 \text{ Mbps}$.

1. Calculate the bandwidth-delay product, $R \cdot d_{\text{prop}}$.
2. Consider sending a file of 800,000 bits from Host A to Host B as one large message. What is the maximum number of bits that will be in the link at any given time?
3. What is the width (in meters) of a bit in the link?

Answer

Given: $R = 500 \text{ Mbps} = 5 \times 10^8 \text{ bps}$, $d_{\text{prop}} = \frac{4.5 \times 10^6}{3 \times 10^8} = 0.015 \text{ s}$.

1. **Bandwidth-delay product:**

$$R \cdot d_{\text{prop}} = 5 \times 10^8 \cdot 0.015 = 7.5 \times 10^6 \text{ bits}$$

2. **Maximum bits in the link:** Since the file size (800,000 bits) is less than the bandwidth-delay product, all 800,000 bits can be in the link.

$$\text{Max bits in link} = 800,000$$

3. **Width of a bit:**

$$\text{Bit width} = \frac{\text{link length}}{\text{bits in link}} = \frac{4.5 \times 10^6}{800,000} \approx 5.625 \text{ m/bit}$$

Question 37

Refer again to problem Q34.

1. How long does it take to send the 50 TB file, assuming it is sent continuously over the 100 Mbps link?
2. Suppose the file is broken into 20 packets of 40,000 bits each. Each packet is acknowledged by the receiver, and the transmission time of an acknowledgment is negligible. The sender cannot send a packet until the preceding one is acknowledged. How long does it take to send the file?
3. Compare the results from (a) and (b).

Answer

1. Continuous transfer over $R = 100$ Mbps:

$$t = \frac{4 \times 10^{14}}{10^8} = 4 \times 10^6 \text{ s} \approx 46 \text{ days}$$

2. Packetized transfer (stop-and-wait, 20 packets of 40,000 bits each): Transmission time per packet:

$$t_{\text{pkt}} = \frac{40,000}{10^8} = 0.0004 \text{ s} \approx 0.4 \text{ ms}$$

Total time: $20 \cdot 0.0004 = 0.008 \text{ s}$ (ignoring propagation for short packets), but practically propagation delay dominates for long-distance links (0.015 s). Using stop-and-wait, total time is $20 \cdot (0.4 \text{ ms} + 0.015 \text{ s}) \approx 0.31 \text{ s}$.

3. **Comparison:** Continuous transfer of the 50 TB file takes days, while stop-and-wait for 20 small packets is negligible. This shows the importance of packet size and protocol overhead; for very large files, continuous high-speed transmission is required.

Question 38

Suppose there is a 10 Mbps microwave link between a geostationary satellite and its base station on Earth. Every minute the satellite takes a digital photo and sends it to the base station. Assume a propagation speed of $s = 2.4 \times 10^8$ m/s.

1. What is the propagation delay of the link?
2. What is the bandwidth-delay product, $R \cdot d_{\text{prop}}$?
3. Let x denote the size of the photo. What is the minimum value of x for the microwave link to be continuously transmitting?

Answer

Given: $R = 10 \text{ Mbps} = 10^7 \text{ bps}$, satellite altitude $h \approx 36,000 \text{ km} = 3.6 \times 10^7 \text{ m}$.

1. Propagation delay:

$$d_{\text{prop}} = \frac{h}{s} = \frac{3.6 \times 10^7}{2.4 \times 10^8} = 0.15 \text{ s}$$

2. Bandwidth-delay product:

$$R \cdot d_{\text{prop}} = 10^7 \cdot 0.15 = 1.5 \times 10^6 \text{ bits}$$

3. Minimum photo size for continuous transmission: To fully utilize the link, the photo size x must be at least the bandwidth-delay product:

$$x_{\text{min}} = R \cdot d_{\text{prop}} = 1.5 \times 10^6 \text{ bits} \approx 187.5 \text{ kB}$$